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23 December 1977

MILITARY SPECIFICATION

# JACK, RAILWAY TRACK, GASOLINE- OR DIESEL-ENGINE-DRIVEN, RAILROAD, WHEEL-MOUNTED

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers hydraulic power jacks (hereinafter called lifter) capable of developing 23,000 pounds (10 400 kg) of lift for track-lifting operations on railroad track, and spotboard equipment.

1.2 Classification. The jacks shall be one of the following types and classes as specified (see 6.2):

Type I - 36-inch (914 mm) Gage.
Type II - 39-3/8-inch (1000 mm) Gage.
Type III - 42-inch (1067 mm) Gage.
Type IV - 56-1/2-inch (1435 mm) Gage.
Type V - 60-inch (1524 mm) Gage.
Class 1 - Gasoline-Engine-Driven.
Class 2 - Diesel-Engine-Driven.

FSC 2230

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| O-E-915 | - Extinguisher, Fire, Dry-Chemical (Hand-Portable). |
| QQ-S-781 | - Strapping, Steel, and Seals. |
| UU-T-81 | - Tags, Shipping and Stock. |
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | - Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | - Boxes, Shipping, Fiberboard. |
| PPP-P-40 | - Packaging and Packing of Hand Tools. |
| PPP-T-60 | - Tape: Packaging, Waterproof. |

MILITARY

| | |
|---|---|
| MIL-P-116 | - Preservation-Packaging, Methods of. |
| MIL-B-121 | - Barrier Material, Greaseproofed, Waterproofed, Flexible. |
| MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |
| MIL-L-2104 | - Lubricating Oil, Internal Combustion Engine, Tactical Service. |
| MIL-P-3321 | - Painting and Marking: Railway Motive Power and Work Equipment. |
| MIL-E-10062 | - Engines: Preparation for Shipment and Storage of. |
| MIL-G-10924 | - Grease, Automotive and Artillery. |
| MIL-G-23827 | - Grease, Aircraft and Instrument, Gear and Actuator Screw. |
| MIL-B-26195 | - Boxes, Wood-Cleated, Skidded, Load-Bearing Base. |
| MIL-A-46153 | - Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package. |
| MIL-L-46167 | - Lubricating Oil, Internal Combustion Engine, Arctic. |



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STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-130 | - Identification Marking of US Military Property. |
| MIL-STD-1186 | - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods. |
| MIL-STD-1410 | - Methods for Selection of Industrial Engines for End Item Application. |
| MIL-STD-1474 | - Noise Limits for Army Materiel. |

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Standard and Recommended Practices.
J512 - Automotive Tube Fittings.
J514d - Hydraulic Tube Fittings.
J516 - Hydraulic Hose Fittings.
J517a - Hydraulic Hose.
J531 - Automotive Pipe, Filler and Drain Plugs.
J532 - Automotive Straight Thread Filler and Drain Plug.
J534 - Lubricating Fittings.
J744 - Hydraulic Power Pumps.
J925 - Minimum Access Dimensions for Construction and Industrial Machinery.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096).



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NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street NW, Washington, DC 20402.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

AMERICAN WELDING SOCIETY, INC.

Standard Qualification Procedure.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Environmental Protection Agency (EPA) Control of Air Pollution From New Motor Vehicles and New Motor Vehicle Engines.

(Application for copies should reference Code of Federal Regulations, 45 CFR, Part 1201 and should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

## 3. REQUIREMENTS

3.1 Description. The railway track lifter shall be portable, gasoline-, or diesel-engine-driven, hydraulic-powered units for lifting railroad track. The track lifter shall be mounted on four ductile-iron or cast-steel,



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single-flanged wheels not less than 6 inches (150 mm) in diameter for movement on track. The lifter shall be designed for one-man operation and shall raise track comprised of rail weighing from 60 to 132 pounds/yard (30 to 65 kg/m).

3.1.1 Drawings. The contractor is responsible for preparing his own engineering drawings. No deviation from the prescribed dimensions is permissible without prior approval of the contracting officer.

3.2 First article (first-produced lifter). The contractor shall furnish a lifter within the time frame specified in the contract, to prove prior to starting production, that his production methods and choice of design detail will produce lifters that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government.

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.10).

3.3.1 Screw threads. All screw threads shall be in accordance with Handbook H28.

3.3.2 Welding. Unless otherwise specified, all welding shall be in accordance with the latest applicable codes of the American Welding Society.

3.4 Interchangeability. Parts and components shall be interchangeable between lifter.

3.5 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, or replacement without the removal of other major assemblies and installed attachments. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage.

3.6 Safety. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions.



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3.6.1 Human factors. The characteristics of the lifter shall provide for operation by personnel ranging from small man (5th Percentile) clothed, through the large man (95th Percentile), arctic-clothed, in accordance with SAE J925. The noise level at the operator's station shall not exceed the limits specified in MIL-STD-1474, Category C, for steady-state noise when tested in accordance with 4.5.2.6 (aural nondetectability and impulse noise do not apply). The precautions of MIL-STD-1474 shall be provided if the sound level exceeds 85 dB(A) when tested in accordance with 4.5.2.6. Hearing protectors shall not be supplied with the lifter. Hazard sign, when required, shall state "CAUTION, HEARING PROTECTION REQUIRED".

3.7 Bearings. All bearings shall be of the antifriction type that will carry the radial and axial loads.

3.8 Braking system. The lifter shall be equipped with a braking system that will hold the lifter in a stationary position when on a 5 percent grade. Wheels shall roll freely when the brake is released.

3.9 Rail clamps. The lifter shall be furnished with clamps which will secure the lifter rigidly to the railroad track, allowing the wheels and frame to remain in firm contact with the track for the power lifting operation. The clamps shall engage the rails before the rams are actuated and shall fit the track for the type of lifter specified including all rail sizes and sections from 60 pound/yard (30 kg/m) class to 132 pound/yard (65 kg/m) class.

3.10 Hydraulic system. The hydraulic system shall provide a means of operating the power lifter. Oil for the system shall conform to MIL-L-2104, Grade 10. For low temperature operation, oil conforming to MIL-L-46167 may be used. All hydraulic component materials shall be compatible with these oils. Unless otherwise specified herein, all system components shall withstand a proof pressure that is equal or greater than 1-1/2 times the maximum working pressure (1-1/2 times relief valve setting) without external leakage, damage, or permanent deformation. The hydraulic system shall be cleaned free of foreign matter (including weld slag and spatter). Hydraulic connection ports of the hydraulic system components shall be sealed by the component manufacturer and this seal shall be maintained until final assembly on the vehicle.

3.10.1 Cylinders. The power lifter shall be provided with two, double-acting hydraulic cylinders (RAMS) fitted with a shoe designed for lowering between railroad ties. The cylinders shall be fabricated from seamless



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steel tubing, or welded and redrawn tubing. All welding and brazing to the cylinder within the zone of piston operation shall be done prior to final machine honing or rolling of the cylinder. All internal cylinder ports shall be located beyond the area of piston travel. Each cylinder shall have an exclusion device at the rod to prevent foreign material and fluids from entering and damaging the seal and other cylinder components. All cylinders shall perform as specified herein without buckling or bending. The travel of the piston rods shall be such that the shoe can be lowered at least 10 inches (255 mm) below the top of the railroad rails and shall have a clearance of not less than 2 inches (50 mm) between the shoes and the rail level when the cylinders are in the raised position. The cylinders shall be located as close as practicable to the track wheels and shall operate from hydraulic pressure supplied by pump(s) powered by the engine. The piston rod shall be hard chrome plated to prevent corrosion. The bearing surface shall be a material that will not scar, score, or gall the inside surface of the cylinder. The lifter shall be provided with a mechanical lock to relieve the pressure on the cylinders and to prevent dropping of the foot jack while in transit or storage.

3.10.2 Pumps. The lifter shall be equipped with hydraulic pump(s) driven directly by a gasoline- or diesel-engine. The pump(s) shall be a constant displacement type and shall be capable of developing the system relief valve setting pressure at a speed corresponding to the manufacturer's rated speed.

3.10.3 Controls. Cylinders shall be provided with individual controls. All hydraulic control valves shall be of the metering type so that accurate control of the unit may be maintained at all times. All controls shall be located within easy reach of the operator when he is in his normal standing or seating position. Pushing of a lever shall result in a lowering of the shoe and pulling shall result in a raising or retracting the shoe. The travel control grip shall have different configuration from that of 1-cylinder control levers. All control-valve spools shall be spring centered with sufficient force to return the levers to the center position and retain it there throughout all operations. Valve spools exposed to dirt and moisture shall be provided with wipers and seals. Spools shall not bind under operating conditions specified herein. Hydraulic control valves shall unload the pump when cylinders or propelling motor are not in use.

3.10.4 Relief valve. A relief valve shall be provided to limit the pump pressure to the maximum required to perform the necessary system functions. Pressure equal to the relief valve setting shall be obtained only when a cylinder is bottomed or when the system is subjected to shock loading.



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3.10.5 Return line filter. The manufacturer's standard hydraulic oil filtration system shall be furnished.

3.10.6 Reservoir. A reservoir shall be furnished to contain the hydraulic system oil. The reservoir shall include a filler opening, screen, cap, and other items required to function as specified herein. The reservoir shall be rigidly mounted to prevent vibrations that may result in leaks, loosening of fasteners, or cracks. Connections on the reservoir shall be located so that connecting lines can be removed without removing the reservoir or any other major component. The reservoir shall be located and constructed so that during operation and performance of all functions specified herein the pump shall not be subject to aeration. All control linkages or connections into the reservoir shall be sealed to prevent leakage. The reservoir should be baffled when necessary to provide a tortuous flow path from the reservoir inlet to the reservoir outlet in order to prevent a direct current between the two ports. A dipstick or sight gage shall be furnished to determine the reservoir oil level. The dipstick shall be marked "Add" and "Full" by etching, stamping, engraving, or embossing. The dipstick shall be removable without tools and shall be located in the filler neck. When practical, the sight gage or window should be visible to the operator when he is in his normal position and shall be marked "Add" and "Full" and should be recessed in the reservoir or protected to prevent breakage. The "Add" and "Full" marks on the sight gage window, or dipstick shall indicate the correct oil level when all hydraulic lines and components are full. Access to the interior of the reservoir shall be provided for repair and removal of all internal parts and components. The filler cap and tube shall be located to provide for filling and to prevent entrance of contaminants throughout filling and checking operations. The reservoir pump suction openings should be located above the bottom of the reservoir to prevent entrance of settled contaminants. Except for sealed reservoir, a breather shall be provided and shall filter incoming air. The breather shall preclude the entrance of water induced by rain.

3.10.7 Hydraulic lines. Hydraulic lines shall consist of tubing or pipe, hose, and fittings. Hose or a combination of hose and tubing shall be used between components that move in relation to each other. All other lines shall be tubing. All lines shall be routed to provide the shortest line, minimum number of bends, ease of maintenance, and maximum protection. Provision shall be made to prevent damage of lines to chafing. The hose installation, such as angle of fittings and location, shall cause no stress concentration on the hose at the fitting, and the hose shall be supported, when necessary, to eliminate sagging at the fitting. All lines other than



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suction lines shall be consided pressure lines. All lines shall be cleaned free of foreign matter including weld slag and spatter.

3.10.7.1 Pressure hose assemblies. Pressure hose shall be as specified in SAE J517 and pressure hose couplings shall be as specified in SAE J516. The hose selected shall have a working pressure which is equal to or greater than the hydraulic system maximum working pressure (relief valve setting). The hose installation shall not cause the hose to be bent beyond the minimum bend radius specified in SAE J517.

3.10.7.2 Pressure tubing and piping. Pressure tubing and piping shall have a minimum burst pressure rating of 3.5 times the system maximum working pressure relief valve setting. All tubing bends shall be smooth without flattening, kinking, or wrinkling of the tube. The term "without flattening" is defined as follows: Throughout the bend, the tube maximum diameter shall be equal to or greater than the tube nominal diameter, and the tube maximum diameter minus the tube minimum diameter shall be not greater than 15 percent of the tube nominal diameter. All tubing and piping shall be cleaned free of foreign matter including cuttings and weld slag and spatter.

3.10.7.3 Pressure pipe fitting and pressure tube fittings. Unless otherwise specified herein, pressure tube fitting shall conform to either SAE J512 or SAE J514. Pressure tube fitting shall have burst pressure rating that is equal to or greater than 3.5 times the maximum system working pressure (relief valve setting).

3.11 Engine. For the purpose of selecting the engine, the lifter shall be classified as a Class I end item in accordance with MIL-STD-1410. The engine shall be gasoline or diesel as specified and shall conform to MIL-STD-1410, Class I unless otherwise specified herein. All systems, components, and accessories shall conform to MIL-STD-1410 unless otherwise specified herein.

3.11.1 Fuel tank. The capacity of the tank shall be sufficient for 8 hours of continuous operation in operating condition specified herein.

3.11.2 Exhaust system. The engine shall be equipped with a muffler and an exhaust pipe located and installed in such a manner that exhaust gases will not be discharged under the jack or be hazardous to operating personnel.

3.11.3 Crankcase ventilation system. A positive crankcase ventilation system in accordance with EPA requirements shall be provided on each engine. The system shall prevent or eliminate emission of crankcase vapors into the atmosphere under all operational conditions specified herein. If the ventilation system is connected to the air cleaner, it shall be connected in such



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a manner that no vapors or condensation will come in contact with, or be deposited on the air cleaner element. The design of the system shall be such that service or replacement to the system, may be accomplished without removal of other components of the engine.

3.12 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.12.1 Lubricants. The lifter, including hydraulic system, shall operate as specified herein when lubricated with military lubricants, or as recommended by the manufacturer for this type of equipment (see 6.6).

3.12.2 Lubrication fittings. Lubrication fittings, when used, shall conform to SAE J534. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun and a 10-inch (254 mm) flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be fastened to provide accessibility to lubrication fittings.

3.12.3 Pressure-release device. A suitable pressure-release device shall be provided where the use of pressure lubricating equipment may damage grease seals or other parts.

3.12.4 Enclosures with integral reservoir of lubricants. Enclosures such as gear cases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch (13 mm) pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with a lubricant and for complete draining. The drain outlet shall be fitted with a magnetic drain-plug conforming to SAE J531 and J532 size 7/8-14UNF-3. Integral tubes or troughs may be used to convey the lubricant from the drain to the ground. Accessibility to the drainplug, the filling means, and the lubricant-level checking device shall be provided without the removal or adjustment of accessories or parts. Plates equipped with hand-operable quick-disconnect fasteners may be removed.

3.12.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as recommended by the manufacturer or as follows:



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(a) MIL-G-23827 - For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250° F (121° C).

(b) MIL-G-10924 - For bearings where operating temperatures do not exceed 175° F (79° C).

Each lifter shall be assembled, run-in (broken-in) for 10 minutes (see 4.5.2.1.), and delivered with these greases. Each component shall be tagged in a conspicuous place to indicate which grease has been used.

3.12.6 Oil lubrication. Unless otherwise preserved for packaging (see 5.2), all components shall be filled to the operating level with winter grades of military or commercial oils designated for use in the temperature range of 0° F to 32° F (-18° to 0° C). Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.13 Fire extinguisher. A 2-1/2-pound (1 kg) fire extinguisher conforming to O-E-915, Type III, Class 2, with mounting bracket, shall be furnished and installed on each lifter. The extinguisher shall be mounted in a position accessible to the operator.

3.14 Lifting attachments. Permanently affixed attachments that enable the lifter to be lifted in its normal travel position shall be provided. Each attachment shall withstand, without damage to any part of the lifter, 2-1/2 times the force imposed when the lifter is suspended by the lifting attachments. The inside diameter of the attachment eye shall be not less than 3 inches (76 mm). Lifting attachments may also be used as tiedown attachments (see 3.15).

3.15 Tiedown attachments. Permanently affixed attachments that permit fastening the lifter to the floor or deck of a transportation medium shall be provided. Attachments shall be rounded and beveled to prevent cutting of the tiedown material. Tiedown attachments may also be used as lifting attachments (see 3.14).

3.16 Identification marking. The lifter shall be identified in accordance with MIL-STD-130. Identification plates shall be photographic plate and shall conform to MIL-P-514, Type I, Style 2. Transportation data plates shall be photographic plate and shall conform to MIL-P-514, Type I, Style 4, and shall indicate location and capacity of lifting and tiedown attachments.



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3.17 Instruction plates. Each lifter shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedure to be followed in operating or servicing the lifter. Instruction plates shall be photographic plate and shall conform to MIL-P-514, Type I, Style 3 or commercial equivalent.

3.18 Treatment and painting. The portions of each lifter and its components and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-P-3321.

3.19 Toolbox. Each lifter shall be provided with a weatherproof toolbox made of steel. The toolbox lid shall be secured to one edge of a side by a continuous hinge extending the full length of the toolbox edge. The lid shall remain in the fully open position either by its own weight or by holding devices. Two spring-tensioned hook-type fasteners shall be provided to keep the lid closed. A hasp which will accomodate a key-operated lock shall be provided for locking the toolbox. A drain hole shall be provided in the bottom of the toolbox. The toolbox shall be of a size to hold the maintenance tools and repair parts specified.

3.20 Lifter spot-board brackets. Type IV lifter shall be fitted with brackets for holding the lifter targets. One bracket shall be located on each side of the machine as close to the rail clamps as possible and spaced so the target may be located over either rail.

3.21 Workmanship.

3.21.1 Steel and aluminum fabrication. Steel and aluminum used in the fabrication of the lifter shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting may be employed instead of sawing and shearing. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.21.2 Bolted connections. Boltholes shall be punched or drilled and shall have the burrs removed. Washers shall be provided in accordance with standard railway practice, and all bolts, nuts, and screws shall be tight.

3.21.3 Riveted connections. Rivet holes shall be punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet holes shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.



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3.21.4 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

3.21.5 Machine work. Tolerance and gages for metal fits shall conform to the limitations specified herein and otherwise within manufacturing tolerances to permit correct fittings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced lifter inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First-produced lifter inspection.

4.3.1 Examination. The first-produced lifter shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first-produced lifter shall be tested as specified in 4.5.2 through 4.5.2.6. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each lifter shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.



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4.4.2 Tests. Each lifter shall be tested as specified in 4.5.2.5. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The lifter shall be examined for the following defects:

101. Wheels for the railway lifter not as specified.
102. Material not as specified.
103. Screw threads not as specified.
104. Welding not as specified.
105. Safety not as specified.
106. Human factors not as specified.
107. Bearings not as specified.
108. Braking system not as specified.
109. Rail clamps not as specified.
110. Hydraulic system not as specified.
111. Cylinders not as specified.
112. Pumps not as specified.
113. Controls not as specified.
114. Relief valve not as specified.
115. Return line filter not as specified.
116. Reservoir not as specified.
117. Hydraulic lines not as specified.
118. Pressure hose assemblies not as specified.
119. Pressure tubing and piping not as specified.
120. Pressure pipe fitting and pressure tube fittings not as specified.
121. Engine not as specified.
122. Fuel tank not as specified.
123. Exhaust system not as specified.
124. Crankcase ventilation system not as specified.
125. Lubrication not as specified.
126. Lubrication fittings not as specified.
127. Pressure-release device not as specified.
128. Enclosures with integral reservoir of lubricants not as specified.
129. Grease lubrication not as specified.
130. Oil lubrication not as specified.
131. Fire extinguisher not as specified.
132. Lifting attachments not as specified.
133. Tiedown attachments not as specified.
134. Identification marking not as specified.
135. Instruction plates not as specified.
136. Treatment and painting not as specified. Toolbox not as specified.



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137. Steel and aluminum fabrication not as specified.
138. Bolted connections not as specified.
139. Riveted connections not as specified.
140. Welding not as specified.
141. Machine work not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to test, the lifter shall be lubricated with oils and greases specified in 3.12 and run-in for 10 minutes exercising all functions without load. Oils shall be those designated for use in the ambient temperature at the place of test. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The test specified in 4.5.2.4 shall be executed by one man. The engine shall be governed at a speed not greater than the continuous rated speed as established by the qualification test. The hydraulic pumps shall not exceed the continuous rpm recommended by the pump manufacturer when operating at not more than 2500 psi (17 250 kPa).

4.5.2.2 Hydraulic system. The hydraulic components, except reservoir, shall be pressurized to not less than 1-1/2 times the maximum working pressure (1-1/2 times relief valve setting) required to meet the requirements of this specification. The reservoir shall be pressurized to 25 psi (170 kPa). Any external leaks, deformation, or breakage of any component shall constitute failure of this test.

4.5.2.3 Lifting. The lifter shall perform the following lifting operations on the applicable gage track: Lift a load of not less than 23,000 pounds (10 430 kg) by lowering the shoe the full extent of ram travel. Hold the load for 30 minutes. After this 30-minute period, the distance of retraction of each ram shall be measured. Inability to raise the required load; any external leaks in the hydraulic system; deformation or breakage of any component shall constitute failure of this test.

4.5.2.4 Performance. Operate the lifter on rail weighing 132 pounds (65 kg) per yard. Stop the lifter on a 5 percent grade and set the brake. Inability of the lifter to remain stationary on a 5 percent grade with the brake set shall constitute failure of this test.

4.5.2.5 Operational test. Each lifter shall be operated for not less than 15 minutes performing all operational functions under specified load. Any operational difficulties, breakage, evidence of contamination on hydraulic filter, leakage, or permanent deformation shall constitute failure of this test.



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4.5.2.6 Noise level measurements. Measure the noise at the operator's station in accordance with MIL-STD-1474 for steady-state noise while operating the lifter. Nonconformance to 3.6.1 shall constitute failure of the test.

4.6 Inspection of packaging.

4.6.1 Preproduction pack inspection.

4.6.1.1 Examination. Examine the preproduction pack for the defects specified in 4.6.2.2. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Test. The preproduction pack of a boxed lifter for Level A shall be subjected to the railroad car test as specified for guided impact tests in accordance with MIL-STD-1186, Appendix A. The car shall strike a string of five empty cars with draft gear extended and the brakes set, at a speed of not less than 10 miles (15 km) or more than 11 miles (20 km) per hour. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the components shall constitute failure of the preproduction pack and shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Examination. Each unit of product shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

142. Preproduction pack not as specified.
143. Preservation not Level A, Level B, or Commercial as specified (see 6.2).
144. Preservation of disassembled parts for Level A not as specified.
145. Preservatives used for Level A not as specified.
146. Unpainted metal surfaces not preserved for Level A as specified.
147. Engine not preserved for Level A as specified.
148. Hydraulic system not preserved for Level A as specified.
149. Small components not preserved for Level A as specified.
150. Preservation of complete lifter for Commercial not as specified.
151. Packing not Level A or Commercial as specified (see 6.2).
152. Each complete lifter not packed for Level A as specified.



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153. Each complete lifter not packed for Commercial as specified.
154. Fire extinguisher not preserved for Level A as specified.
155. Repair parts not preserved for Level A as specified.
156. Technical publications not preserved for Level A as specified.
157. Maintenance tools not preserved for Level A as specified.
158. Marking for military levels of protection not in accordance with MIL-STD-129.
159. Marking for commercial level of packaging not as specified.

## 5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination and test of the boxed lifter within the time frame specified (see 6.2) to prove prior to starting production that the applied preservation, packing, and marking comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be accomplished utilizing either the preproduction lifter or the production lifter. If the preproduction model lifter is utilized, any preservation, and packing shall be removed by the contractor at no expense to the Government when requested by the Government to facilitate comparison between the preproduction model lifter and the production lifter.

5.2 Preservation. Preservation shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly. Disassembly of the lifter shall be the minimum necessary to safeguard parts vulnerable to damage or loss and to accomplish reduction in cube, in accordance with the manufacturer's recommendations. Removed bolts, nuts, screws, washers, and pins shall be reinstalled in mating parts to prevent their loss. Removed keys shall be secured to the keyway of the primary component or shaft.

5.2.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.3 Unpainted metal surfaces. Unpainted exterior metal surfaces not specified herein requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative.



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5.2.1.4 Engine. The engine and engine accessories shall be preserved in accordance with MIL-E-10062, Type II, Level A, Method I. If a liquid-cooled engine is furnished, the cooling system shall be preserved by the water and antifreeze procedure, except the antifreeze shall conform to MIL-A-46153.

5.2.1.5 Hydraulic system. The hydraulic fluid supply reservoir shall be filled with the hydraulic fluid recommended for operation. The pistons shall be retracted as far as practicable into the cylinders and secured. Any exposed portions of the piston rods shall be coated with Type P-6 or P-11 preservative and wrapped or covered with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2. The wrap shall be secured in place with tape conforming to PPP-T-60, Type IV. When the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with Type P-1 preservative, with no wrapping required. The hydraulic control valves shall be secured in the neutral position and preserved as specified for the piston rods. Hydraulic lines shall not be disconnected. A tag conforming to UU-T-81, Type A, shall be attached in a conspicuous location indicating: "The hydraulic system is filled with the fluid recommended for operation." The information shall be applied on the tag with water-proof ink.

5.2.1.6 Fire extinguisher. Unpainted exterior surfaces of the fire extinguisher requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with Type P-1 preservative. The extinguisher shall then be placed in a close-fitting box conforming to PPP-B-636, W6c. The box shall be closed and secured in accordance with Method V of the appendix to the box specification.

5.2.1.7 Repair parts. The preservation application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve the repair parts.

5.2.1.8 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3 and placed in the toolbox.

5.2.1.9 Maintenance tools. Maintenance tools shall be preserved in accordance with the Level A requirements of PPP-P-40 and placed in the toolbox.

5.2.1.10 Consolidation. Small loose components that cannot be placed in the toolbox and that require the protection afforded by a box to prevent damage, pilferage, or loss shall be placed together in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional, Grade B, or PPP-B-621,



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Class 2, style optional. The boxes shall be of a size most convenient for securing to the lifter or to the base of the box specified in 5.3.1 without increasing cube. The contents shall be blocked, braced, and anchored or cushioned as applicable to prevent free movement or damage. Box closure shall be in accordance with the appendix to the applicable box specification. Strapping shall not be required. Large components shall be secured to the lifter as provided for, or to the base specified in 5.3.1 with zinc coated strapping conforming to QQ-S-781 size as applicable, in a manner to prevent movement or damage.

5.2.2 Level B. Each complete lifter shall be preserved as specified for Level A, with the following exceptions.

5.2.2.1 Consolidation. Boxes for consolidation shall be Domestic Type or Class as applicable and strapping shall not be required to be zinc coated.

5.2.3 Commercial. Each complete lifter and all items furnished with the lifter shall be preserved to afford protection against mechanical and physical damage. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coatings, volatile corrosion inhibitors or desiccated packs. Items requiring protection from physical damage shall be protected by wrapping, or cartonizing of the individual item. If a liquid cooling system is furnished, it shall be filled with a fresh clean solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153, and a tag attached indicating the temperature to which the cooling system can be subjected before damage. All lubrication reservoirs, sumps, etc., shall be filled to the operating level with the grade of military lubricant designated for use in the temperature range at the shipping destination. A tag shall be attached in a conspicuous location to indicate which lubricant has been used, the temperature range of lubricant used, and grade of lubricant used. A deprocessing guide shall be furnished indicating in detail the preservation material to be removed and any other servicing required prior to placing the lifter in operation.

5.3 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

5.3.1 Level A. Each complete lifter, preserved as specified in 5.2 shall be packed in a box conforming to MIL-B-26195, Type II, Style C, Class 2. Rubbing strips shall be applied to the skids. Superstructure components shall be of plywood. Blocking, bracing, and anchoring of the contents shall be in accordance with MIL-STD-1186. Box closure and strapping application shall be in accordance with PPP-B-601. Strapping shall be zinc coated.



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5.3.2 Commercial. Each complete pack, preserved as specified in 5.2, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings. Containers shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking.

5.4.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Marking for commercial packaging shall reflect as a minimum, the item name, National Stock Number, quantity, contract number, and appropriate address markings. The marking shall be applied with a waterproof material, be legible, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The railway track lifter is intended for use in raising track for surfacing and timbering railroad track.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of lifter required (see 1.2).
(c) When spot-board equipment is required (see 6.3).
(d) Time frame required for submission of the preproduction pack (see 5.1).
(e) Degree of preservation and packing required (see 5.2 and 5.3).

6.3 Spot-board equipment. When specified, spot-board equipment Type IV lifter shall be furnished and stored on the lifter.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the lifter in accordance with this specification.

6.5 Government serial numbers. The contracting officer should furnish the Government serial numbers for the lifter upon award of the contract or order.

6.6 Lubricants. The contracting officer should furnish a list of lubricants applicable to the lifter covered by this specification with the commercial equivalent when requested by the contractor.



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6.7 First-produced jacks. Any changes or deviations of production lifter from the approved first-produced lifter during production will be subject to the approval of the contracting officer. Approval of a first-produced lifter will not relieve the contractor of his obligation to furnish lifter conforming to this specification.

6.8 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operational manual to be furnished with each lifter.

6.9 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary including any special tools and instructions on shipment of lifter.

6.10 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

Custodians:
Army - ME

Preparing activity:
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Project 2230-A047



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-J-14158E(ME) Jack, Railway Track, Gasoline- or Diesel-Engine-Driven, Railroad, Wheel-Mounted

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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